

42.1 Polynomial regression and its instability

The simplest way to let a regression curve bend is to add powers of the regressor as columns. Nothing in Parts II and III changes, because nothing there ever asked what the columns *were*.

Definition 42.1.1. Polynomial regression

Let $x_1, \dots, x_n$ be observed values of a single regressor. The **polynomial regression model of degree d** is

$$E(Y_i) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d, \quad i = 1, \dots, n,$$

a linear model with the $n \times (d+1)$ **Vandermonde** matrix $\mathbf{X} = [\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots, \mathbf{x}^d]$, where $\mathbf{x}^j$ has entries x_i^j . More generally, any functions $\phi_0, \dots, \phi_d$ give columns $\phi_j(\mathbf{x})$ and a fitted curve $\hat{f}(t) = \sum_j \hat{\beta}_j \phi_j(t)$; they are a **basis**, and the model is the space $\mathcal{P}_d = \text{span}\{\phi_0, \dots, \phi_d\}$, not the list.

42.1.1 Everything from Parts II and III applies

Write $\mathbf{X}$ for the Vandermonde matrix. Then $\hat{\boldsymbol{\beta}}$ is unbiased as soon as the x_i take $d+1$ distinct values ([Exercise \(A1\)](#)); the fitted curve $\hat{f}(t) = \mathbf{x}(t)^\top \hat{\boldsymbol{\beta}}$, with $\mathbf{x}(t) = (1, t, \dots, t^d)^\top$, is estimable at every t , so Theorem 12.1.1 gives it an interval and Theorem 12.4.1 a band; raising the degree is a nested comparison (Theorem 11.1.1); replicated x values give the lack-of-fit test of Theorem 14.6.2 against the pure error of Definition 14.6.1; and the diagnostics of [Chapter 20](#) are untouched. By Theorem 6.7.1 none of this depends on the basis, only on $\mathcal{P}_d$.

Warning. What does *not* carry over is the reading of a coefficient: there is no way to change x while holding x^2 fixed, so nothing beyond the intercept has the “other regressors held fixed” meaning of Theorem 6.6.2. Only the curve and its contrasts $f(t_2) - f(t_1)$ mean anything, and whether those describe an intervention is the question of [Chapter 25](#).

42.1.2 What the degree buys

Raising the degree adds a column, so bias falls and variance rises: the trade-off of Example (Choosing a polynomial degree). The useful question is how the two move in practice.

Example. Degree in the carbon dioxide series

The monthly mean carbon dioxide at Mauna Loa, March 1958 to December 2001, gives $n = 521$ observations; Example (A curved trend and a seasonal cycle) modelled it with a quadratic trend and two seasonal harmonics. Keep the four harmonic columns and replace the quadratic by a polynomial of degree d in time rescaled to $[-1, 1]$. The residual standard deviation falls from 0.7602 ppm at $d = 2$ ($p = 7$) to 0.4874 at $d = 8$ and 0.4300 at $d = 16$, while no harmonic coefficient moves by more than 0.0150 ppm.

The improvement is real: panel (a) of [Figure 42.1.1](#) shows the residuals of the quadratic fit wandering by a part per million over decades, and the higher-degree trends tracking that wandering. The price is in panel (b). In the middle of the range the standard error of the fitted mean is much the same at all three degrees — 0.0830, 0.0676, 0.0730 ppm at 1980 —

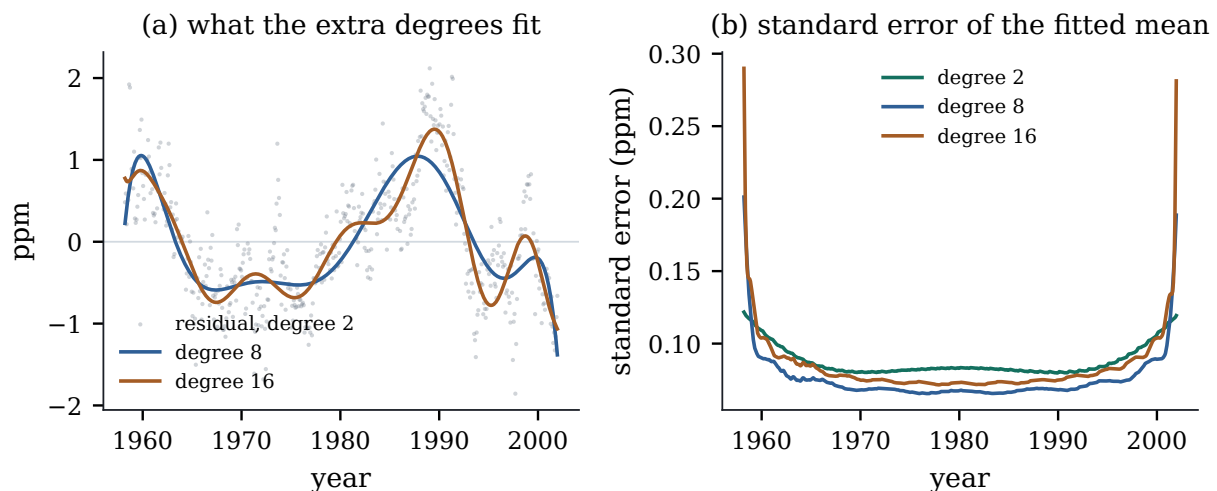


Figure 42.1.1: Polynomial trends for the Mauna Loa series, seasonal columns fixed. (a) Residuals of the quadratic fit, with the degree-8 and degree-16 trends measured from it. (b) Standard error of the fitted mean.

but at the left-hand edge it is 0.1217, 0.2011, 0.2899, so the ratio of edge to centre rises from 1.47 to 2.97 to 3.97. Leverages tell the same story: average 0.0134, 0.0250, 0.0403 against maximum 0.0256, 0.1702, 0.4546. At degree 16 one month, the first of the series, carries almost half of its own fitted value.

```
import numpy as np
import statsmodels.api as sm

co2 = sm.datasets.co2.load_pandas().data["co2"]
monthly = co2.resample("MS").mean().dropna()           # weekly values -> monthly
means
y = monthly.to_numpy()
t = (monthly.index.year + (monthly.index.month - 0.5) / 12).to_numpy()
n = len(y)

def harmonics(s):
    """The seasonal columns of Example 5.3, held fixed throughout the chapter."""
    return np.column_stack([np.cos(2 * np.pi * s), np.sin(2 * np.pi * s),
                             np.cos(4 * np.pi * s), np.sin(4 * np.pi * s)])

lo, hi = t.min(), t.max()
u = lambda s: 2 * (s - lo) / (hi - lo) - 1           # time rescaled to [-1, 1]

print(f"n = {n}, time from {lo:.2f} to {hi:.2f}")
```

[Run this code online](#)

```
def trend_design(s, d):
    """Model matrix: powers 1, u, ..., u^d of rescaled time, then the four harmonics
    ."""
    return np.column_stack([np.vander(u(s), d + 1, increasing=True), harmonics(s)])

for d in (2, 8, 16):
```

```

X = trend_design(t, d)
beta, *_ = np.linalg.lstsq(X, y, rcond=None)
resid = y - X @ beta
p = X.shape[1]
lev = np.einsum("ij,ij->i", X @ np.linalg.pinv(X.T @ X), X) # leverages h_ii
print(f"degree {d:2d}: p = {p:2d} sigma_hat = {np.sqrt(resid @ resid / (n - p)):.4f}")
f" mean h = {lev.mean():.4f} max h = {lev.max():.4f}"

```

[Run this code online](#)

42.1.3 Four ways a polynomial misbehaves

The pattern in [Example \(Degree in the carbon dioxide series\)](#) is not particular to that series: it is the visible part of four properties of the monomial basis and of the polynomial class itself.

Proposition 42.1.1. Instability of a high-degree polynomial fit

Let $\mathbf{X}_d = [\mathbf{1}, \mathbf{x}, \dots, \mathbf{x}^d]$ be the Vandermonde matrix of a design with at least $d + 1$ distinct points, and let $\hat{f}_d$ be the least squares polynomial of degree d .

- (a) (*Conditioning.*) If the design points are a sample from a distribution on $[0, 1]$ with density g , then $n^{-1}\mathbf{X}_d^\top \mathbf{X}_d \rightarrow \mathbf{G}$ almost surely, where $G_{jk} = \int_0^1 x^{j+k} g(x) dx$. For the uniform density this limit is the $(d + 1) \times (d + 1)$ Hilbert matrix $G_{jk} = 1/(j + k + 1)$, whose condition number grows exponentially in d .
- (b) (*Oscillation.*) Interpolation by a polynomial of degree $n - 1$ at n equally spaced points need not converge as n grows, even for a function that is analytic on the interval; the error can diverge. The failure is a property of the node placement, not of polynomials: at Chebyshev nodes the same interpolants converge geometrically.
- (c) (*Extrapolation.*) Let $\tilde{\mathbf{x}}_d = (\mathbf{I} - \mathbf{M}_{d-1})\mathbf{x}^d$ be the part of the top power orthogonal to the lower ones. Then $t \mapsto \text{Var}\{\hat{f}_d(t)\}$ is a polynomial of degree exactly $2d$ with leading coefficient $\sigma^2/\|\tilde{\mathbf{x}}_d\|^2$, so

$$\frac{\text{Var}\{\hat{f}_d(t)\}}{\sigma^2 t^{2d}} \rightarrow \frac{1}{\|\tilde{\mathbf{x}}_d\|^2} \quad \text{as } |t| \rightarrow \infty.$$

Each extra degree therefore multiplies the standard error of the extrapolated curve by a further factor $|t|$, up to the constant $\|\tilde{\mathbf{x}}_d\|/\|\tilde{\mathbf{x}}_{d+1}\|$.

- (d) (*Non-locality.*) Changing one response y_i by Δ changes the fitted curve at every t by $\Delta \ell_i(t)$, where $\ell_i(t) = \mathbf{x}(t)^\top (\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{x}_i$ is itself a polynomial of degree at most d and never identically zero, so it vanishes at most d times: no observation can be given a purely local influence.

Proof.

- (a) The (j, k) entry of $n^{-1}\mathbf{X}_d^\top \mathbf{X}_d$ is $n^{-1} \sum_i x_i^{j+k}$, an average of bounded independent and identically distributed variables, which by the strong law converges almost surely to $E(X^{j+k}) = \int x^{j+k} g$; for $g \equiv 1$ that is $1/(j + k + 1)$. The exponential growth of $\kappa(\mathbf{G})$ is classical and not reproduced here: Todd (1954) shows it is of order $(1 + \sqrt{2})^{4d}$ up to algebraic factors, and Higham (2002) treats the Hilbert matrix and Vandermonde systems. By Eq. 10.1.2 the normal equations (Theorem 6.4.2) then lose about $2 \log_{10} \kappa(\mathbf{X}_d)$ of

the sixteen digits of double precision.

- (b) This is Runge's phenomenon, stated here without proof: Runge (1901) gave the example and Trefethen (2019) the modern account, Chebyshev nodes included. Nothing later depends on the divergence itself, only on the weaker statement that a polynomial forced to follow the data in one region must do something in the others, and that something is an oscillation largest at the ends.

- (c) With $\mathbf{x}(t) = (1, t, \dots, t^d)^\top$,

$$\begin{aligned}\text{Var}\{\hat{f}_d(t)\} &= \sigma^2 \mathbf{x}(t)^\top (\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{x}(t) \\ &= \sigma^2 \sum_{j=0}^d \sum_{k=0}^d [(\mathbf{X}_d^\top \mathbf{X}_d)^{-1}]_{jk} t^{j+k},\end{aligned}$$

a polynomial in t whose coefficient of t^{2d} is $\sigma^2[(\mathbf{X}_d^\top \mathbf{X}_d)^{-1}]_{dd}$. By Eq. 6.6.1 with $\mathbf{X}_1 = \mathbf{x}^d$ and $\mathbf{X}_2 = \mathbf{X}_{d-1}$, that entry is $1/\|(\mathbf{I} - \mathbf{M}_{d-1})\mathbf{x}^d\|^2 = 1/\|\tilde{\mathbf{x}}_d\|^2$, which is positive because $\mathbf{x}^d \notin \mathcal{C}(\mathbf{X}_{d-1})$ when the design has at least $d+1$ distinct points (Exercise A1). A polynomial with positive leading coefficient c satisfies $q(t)/t^{2d} \rightarrow c$, which is the claim.

- (d) Least squares is linear in $\mathbf{y}$, so the fitted curve changes by $\Delta \mathbf{x}(t)^\top (\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{x}_i$ when y_i changes by Δ . In t this is a combination of $1, t, \dots, t^d$, hence a polynomial of degree at most d , and never the zero polynomial, since $(\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{x}_i = \mathbf{0}$ would force $\mathbf{x}_i = \mathbf{0}$ while $\mathbf{x}_i$ has leading entry one. Such a polynomial has at most d real roots. ■

Part (a) explains the numbers in Example (Polynomial regression), and part (b) is drawn in panel (a) of Figure 42.1.2. Part (d) is why a polynomial fit is never local: moving the December 2001 carbon dioxide value by 10 ppm moves the fitted trend by 0.170 ppm at the end of the series and 0.059 ppm in March 1958 when $d = 2$, and by 2.749 and 0.197 ppm when $d = 12$, the degree-12 weight changing sign 12 times on the way.

Example. Runge's function

Interpolate $f(x) = 1/(1 + 25x^2)$ at m equally spaced points of $[-1, 1]$ by the polynomial of degree $m - 1$. The largest error on the interval is 1.92 for $m = 11$, 59.82 for $m = 21$ and 2388.03 for $m = 31$: the function is smooth and bounded by one, and the interpolant does not converge to it. Move the same nodes to the Chebyshev positions $\cos\{(2k + 1)\pi/2m\}$, which crowd towards the ends, and the errors become 0.10915, 0.01533, 0.00206. Only where the information sits has changed.

Trouble at the ends is a shortage of design points relative to what the fitted function is asked to do there. Either add points, rarely an option, or ask the function to do less — which is what knots and penalties do.

Example. Eleven years out of sample

Fit the same models to the 389 months up to the end of 1990 and predict the 132 months of 1991–2001. Inside the fitting range the residual standard deviation falls steadily with the degree — 0.439, 0.428, 0.417, 0.406 ppm at degrees 2, 4, 6, 8 — while outside it the root mean squared prediction errors are 3.99, 2.63, 13.92 and 188.23 ppm. The degree-8 fit, the best of the four in-sample, is out by -540.7 ppm in December 2001 against an observed value

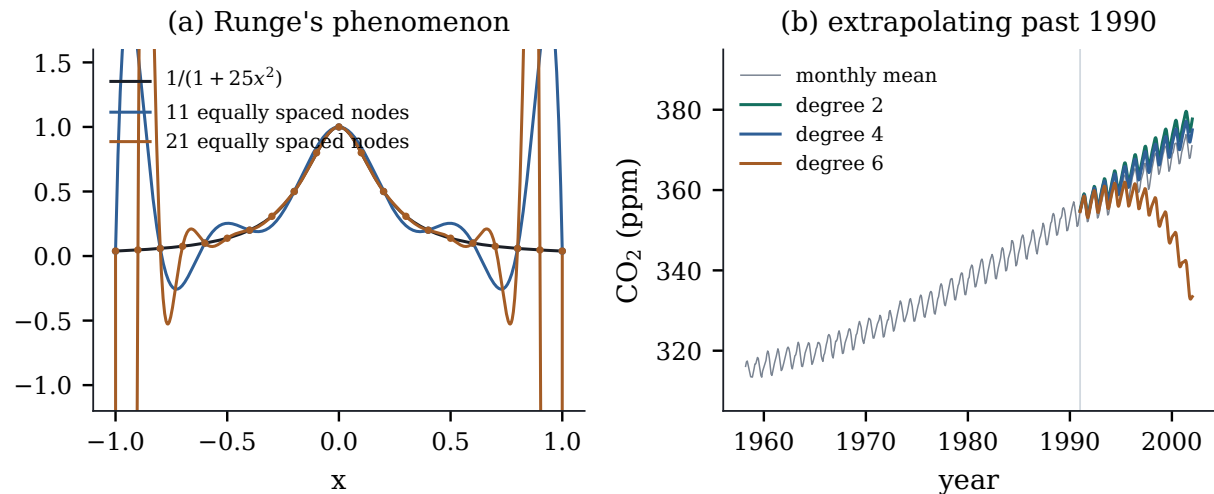


Figure 42.1.2: (a) Runge's phenomenon: equally spaced interpolation of $1/(1 + 25x^2)$, 11 and 21 nodes. (b) Polynomial trends fitted up to 1990, continued to 2002.

near 371: it predicts a concentration several hundred parts per million below zero.

```
import numpy as np

runge = lambda x: 1.0 / (1.0 + 25.0 * x ** 2)
grid = np.linspace(-1.0, 1.0, 2001)

for m in (5, 11, 21, 31):
    equal = np.linspace(-1.0, 1.0, m)          # m equally spaced
    nodes                                     nodes
    cheb = np.cos((2 * np.arange(m) + 1) * np.pi / (2 * m))  # m Chebyshev nodes
    err = [np.max(np.abs(np.polyval(np.polyfit(z, runge(z), m - 1), grid) - runge(
    grid)))
           for z in (equal, cheb)]
    print(f"{m:2d} nodes, degree {m - 1:2d}:  equally spaced {err[0]:10.4f}
    Chebyshev {err[1]:8.5f}")
```

[Run this code online](#)

42.1.4 Centring and scaling are a partial repair

Part (a) of [Proposition 42.1.1](#) is about the *basis*, and a basis can be changed without touching the model. The cheapest change is affine: fit in $u = \{2x - \max x - \min x\}/(\max x - \min x)$, which runs over $[-1, 1]$, or at least in $x - \bar{x}$. By Theorem 6.7.1 the fit is identical and only the arithmetic differs. For the carbon dioxide design the condition number of the trend columns is

Degree	raw years	centred	rescaled to $[-1, 1]$
2	1.08×10^{11}	3.19×10^2	3.76×10^0
4	1.36×10^{22}	1.42×10^5	1.90×10^1
8	9.87×10^{40}	3.11×10^{10}	5.49×10^2
12	1.48×10^{54}	7.17×10^{15}	1.69×10^4
16	2.63×10^{67}	3.25×10^{21}	5.38×10^5

Centring gains nine orders of magnitude at degree 2 and forty-six at degree 16, because raising a number near 1980 to the sixteenth power is what made the columns incommensurable; rescaling gains a further two to sixteen, close to the best any diagonal rescaling can do (Proposition 10.5.2). But the rescaled design is still at 5.38×10^5 by degree 16 and grows geometrically after that, high powers of a number in $[-1, 1]$ all looking like a spike at the ends. The next section removes that dependence exactly.

```
for d in (2, 4, 8, 12, 16):
    raw = np.linalg.cond(np.vander(t, d + 1, increasing=True))
    centred = np.linalg.cond(np.vander(t - 1980.0, d + 1, increasing=True))
    scaled = np.linalg.cond(np.vander(u(t), d + 1, increasing=True))
    print(f"degree {d:2d}:  raw {raw:.2e}  centred {centred:.2e}  scaled {scaled:.2e}")
```

[Run this code online](#)

Two complaints have now been separated. *The basis is bad* is a numerical statement, fixed exactly by a change of basis that no fitted value feels: [Section 42.2](#). *The family is bad* is a statement about $\mathcal{P}_d$ itself, which only a different space of functions can fix: [Section 42.3](#).

42.1.5 Exercises

42.1.5.1 A. Check your understanding

Exercise. A1

Show that $\mathbf{X}_d = [\mathbf{1}, \mathbf{x}, \dots, \mathbf{x}^d]$ has full column rank if and only if the x_i take at least $d + 1$ distinct values, and deduce that a polynomial of degree $n - 1$ through n points with distinct x_i exists, is unique and interpolates exactly.

Solution. If $\mathbf{X}_d \mathbf{c} = \mathbf{0}$ then $q(t) = \sum_j c_j t^j$, of degree at most d , vanishes at every distinct design value; with $k \geq d + 1$ of them a nonzero q would have more roots than its degree, so $\mathbf{c} = \mathbf{0}$. If $k \leq d$, the nonzero $q(t) = \prod_{l \leq k} (t - x_{(l)})$ vanishes on the design. Taking $d = n - 1$ makes $\mathbf{X}$ square of full rank, so the residuals vanish.

42.1.5.2 B. Practice

Exercise. B1

For the design $x_i = (i - 1)/(n - 1)$, show that the (j, k) entry of $n^{-1} \mathbf{X}_d^\top \mathbf{X}_d$ converges to $1/(j + k + 1)$ for each fixed $j, k \leq d$: part (a) of [Proposition 42.1.1](#) for a fixed rather than a random design.

Exercise. B2

For the design $x_i = -1 + 2(i - 1)/(n - 1)$ with $n = 50$, compute the leverage of the leftmost point for $d = 1, 3, \dots, 15$ and plot it against d , comparing with the average $(d + 1)/n$ and the bound of Proposition 6.8.1.

Exercise. B3

Let $\mathbf{u} = a\mathbf{x} + b\mathbf{1}$ with $a \neq 0$. Show that $\mathcal{C}([\mathbf{1}, \mathbf{u}, \dots, \mathbf{u}^d]) = \mathcal{C}([\mathbf{1}, \mathbf{x}, \dots, \mathbf{x}^d])$ and that the change of basis matrix is triangular with diagonal entries a^j . Deduce that centring and scaling leave $\hat{\mathbf{Y}}$, $\hat{\boldsymbol{\varepsilon}}$ and every leverage unchanged, and say why they nevertheless change $\kappa(\mathbf{X})$.

Solution. Expanding $(ax + b)^j$ writes $\mathbf{u}^j$ in terms of $\mathbf{1}, \mathbf{x}, \dots, \mathbf{x}^j$ with leading coefficient a^j , so $\mathbf{T}$ is upper triangular with $T_{jj} = a^j \neq 0$, hence nonsingular, and the column spaces agree. By Theorem 6.7.1 $\mathbf{M}$ is unchanged, so fitted values, residuals and leverages are too. The condition number is a property of $\mathbf{X}$, not of $\mathcal{C}(\mathbf{X})$.

Exercise. B4

Verify part (d) of [Proposition 42.1.1](#) numerically: for the degree-12 trend on the carbon dioxide design compute $\ell_i(t)$ for the last observation and count its sign changes over the data, then repeat for $d = 2$. Why is the count for $d = 2$ smaller than 2 in some designs?

Solution. The counts are 12 and 2. A polynomial of degree d has at most d real roots but may have fewer, and roots outside the range of the data are not counted: if the parabola's vertex lies outside $[\min x, \max x]$, only one sign change is visible.

42.1.5.3 C. Going deeper**Exercise. C1**

The Weierstrass approximation theorem says that every continuous f on $[a, b]$ is a uniform limit of polynomials. Using parts (a) and (c) of [Proposition 42.1.1](#), explain why this is not an argument for fitting high-degree polynomials to data. What does the theorem guarantee about the bias of the degree- d fit as d grows, and what about the variance?

Exercise. C2

For interpolation at nodes $x_1 < \dots < x_m$ the interpolant is $\sum_k f(x_k)L_k(t)$, with L_k the Lagrange basis, and the **Lebesgue constant** is $\Lambda = \max_t \sum_k |L_k(t)|$. Show that the interpolation error is at most $(1 + \Lambda)$ times that of the best uniform approximation of degree $m - 1$, compute Λ for $m = 5, 9, 13, 17$ equally spaced and Chebyshev nodes, and relate the two sequences to [Example \(Runge's function\)](#).